

Lecture 5: American Derivatives

September 22, 2010

1 Introduction

We study American option contracts in the Lectures 5 and 6. Different from European put/call contracts which have only one payment at maturity, **American** contract can be exercised to receive payment at any time up to and including maturity. Because of the optionality of exercising it, American contract is worth no less than its European counterpart. The value of this early exercise feature is called the **early exercise premium**.

Due to the optionality of American contracts, its price is not just the discounted expectation of the payoff at maturity. This is because, the holder of an American contract can decide whether to exercise it now to receive a payment, or not to do so. The holder will compare the value of the contract if he exercise it now, with the discounted expected value of the contract if he chooses to hold it for one more period. When pricing an American contract, the optionality for the holder has to be taken into consideration.

We will see later that, there is a optimal way of choosing the exercise time for an American contract. This is a stopping rule, which can be implemented. That is, the stopping rule can tell us if we need to exercise the contract now or not based on history. After this optimal exercise time, the discounted value of the contract will decrease on average - that is why the exercise time is optimal. As a result, the discounted value of an American contract is not a martingale under the risk-neutral measure. In probability theory, we say the stochastic process $X = \{X_k\}$ is a **supermartingale** if this process has decaying conditional expectation:

$$X_k \geq \tilde{E}(X_{k+1}|\mathcal{F}_k) \geq \tilde{E}(X_m|\mathcal{F}_k), \text{ for } m > k.$$

So the discounted value process of an American contract is actually a supermartingale.

We will see that, similar as European contracts, the value process of an American contract in a N -period binomial model can be fully replicated with a portfolio of stocks and cash until the maturity of this contract. However, the discounted value process of a portfolio with just stocks and cash is a martingale¹. To reflect the fact that the discounted expected value of the contract is actually decreasing with time, we need to withdraw money with the portfolio if necessary.

The optimal stopping rule for exercising an American contract will be discussed next time.

2 Path-independent American derivatives

In the classical N -period binomial model, consider an American derivative that has a payoff function g . That is, if the holder of the derivative exercise it at time $1 \leq k \leq N$, then he will receive $g(S_k)$ from the contract. Let us denote by $V_k(S_k)$ the price of this contract, then there must be

$$V_k(S_k) \geq g(S_k). \quad (1)$$

The value $g(S_k)$ is called the intrinsic value of the contract at time k .

The price V_k can be evaluated using backward induction:

- At time N , if $S_N = s$, then the holder has the right to exercise the contract to receive $g(s)$ or not to exercise it and receive zero. Thus

$$V_N(s) = \max(g(s), 0). \quad (2)$$

¹We have seen this in Chapter 2.

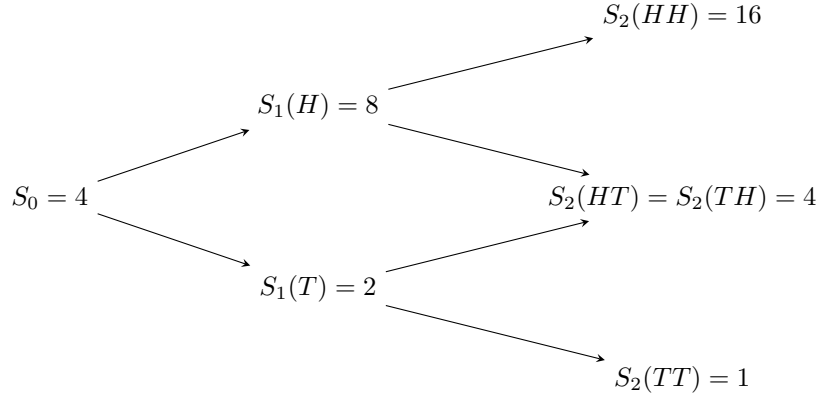


Figure 1: A two-period binomial model.

- If V_{k+1} has been computed, for $k = N - 1, N - 2 \dots$, then at time k , if $S_k = s$, the holder has the right to exercise the contract now to receive $g(s)$, or to keep this contract and expect a payoff that is equal to $V_{k+1}(us)$ or $V_{k+1}(ds)$ at time $k + 1$. Thus

$$V_k(s) = \max \left(g(s), \frac{1}{1+r} [\tilde{p}V_{k+1}(us) + \tilde{q}V_{k+1}(ds)] \right). \quad (3)$$

Example 1. In the two-period model of Figure 1, let the interest rate be $\frac{1}{4}$, so the risk-neutral probabilities are $\tilde{p} = \tilde{q} = \frac{1}{2}$. Consider an American put option, expiring at time two, with strike price 5. Then the fair price V_k can be computed as follows.

$$V_2(s) = (5 - s)^+, \quad (4)$$

$$V_k(s) = \max \left((5 - s)^+, \frac{2}{5} [V_{k+1}(2s) + V_{k+1}(\frac{1}{2}s)] \right), \quad k = 1, 0. \quad (5)$$

In particular,

$$\begin{aligned} V_2(16) &= 0, \quad V_2(4) = 1, \quad V_2(1) = 4, \\ V_1(8) &= \max \left(0, \frac{2}{5}(0 + 1) \right) = 0.40, \quad V_1(2) = \max \left(3, \frac{2}{5}(1 + 4) \right) = 3, \\ V_0(4) &= \max \left(1, \frac{2}{5}(0.40 + 3) \right) = 1.36. \end{aligned}$$

To replicate the value of the put option, we setup a portfolio with $X_0 = 1.36$ at time zero, and long Δ_0 shares of stock, so that if H shows up in the first toss, we have

$$0.40 = V_1(S_1(H)) = \Delta_0 S_1(H) + (1+r)(X_0 - \Delta_0 S_0) = 8\Delta_0 + \frac{5}{4}(1.36 - 4\Delta) = 3\Delta_0 + 1.70.$$

Similarly, if T shows up, we must have

$$3 = V_1(S_1(T)) = \Delta_0 S_1(T) + (1+r)(X_0 - \Delta_0 S_0) = 2\Delta_0 + \frac{5}{4}(1.36 - 4\Delta) = -3\Delta_0 + 1.70.$$

Therefore,

$$\Delta_0 = \frac{V_1(8) - V_1(2)}{8 - 2} = \frac{0.40 - 3}{8 - 2} = -0.43.$$

If the first toss results in a H , then we need to choose a $\Delta_1(H)$ so that

$$\begin{aligned} 0 &= V_2(S_2(HH)) = \Delta_1(H)S_2(HH) + (1+r)(X_1 - \Delta_1(H)S_1(H)) = 16\Delta_1(H) + \frac{4}{5}(0.40 - 8\Delta_1(H)), \\ 1 &= V_2(S_2(HT)) = \Delta_1(H)S_2(HT) + (1+r)(X_1 - \Delta_1(H)S_1(H)) = 4\Delta_1(H) + \frac{4}{5}(0.40 - 8\Delta_1(H)), \end{aligned}$$

Therefore

$$\Delta_1(H) = \frac{V_2(16) - V_2(4)}{16 - 4} = -\frac{1}{12}.$$

In a N -period binomial model, the above algorithm can be implemented for pricing and hedging. And the number Δ_k is given by

$$\Delta_k = \frac{V_{k+1}(uS_k) - V_{k+1}(dS_k)}{(u - d)S_k}. \quad (6)$$

The replication has some subtle points. Let us define

$$C_k = V_k(S_k) - \frac{1}{1+r}[\tilde{p}V_{k+1}(uS_k) + \tilde{q}V_{k+1}(dS_k)]. \quad (7)$$

Because of (3), we have that $C_k \geq 0$. We setup a portfolio with initial cost $X_0 = V_0(S_0)$ and buy Δ_0 shares of stock, and define recursively forward in time the portfolio values $X_1, X_2, \dots, X_N$ by

$$X_{k+1} = \Delta_k S_{k+1} + (1+r)(X_k - C_k - \Delta_k S_k), \quad (8)$$

then we will always have²

$$X_k = V_k(S_k), \quad k = 0, 1, \dots, N. \quad (9)$$

In words, we need to withdraw C_k from X_k at time k , then purchase Δ_k shares of stock and deposit the rest into the money market. This portfolio will replicates the value process of the American derivative. In most cases, C_k will be zero initially, so the discounted value process of the replicating portfolio is a martingale initially. When there is a k such that $C_k > 0$, we know that the optimal exercise time has already past.

We need a formal treatment for this optimal exercise time.

3 Stopping Times

In order to formally characterize the optimal exercise time in probability theory, we need the notion of stopping time. Stopping time is one kind of random variables that can only take values in $\{0, 1, 2, \dots, N\} \cup \{\infty\}$ (in a N -period binomial model). In this section, we will denote stopping times by τ . The random variable τ is random, however, we do not want it to be too random. In all, a stopping time is a rule that tells us when to stop and exercise, the decision for stopping should be made based on just observations that are already made. That is, based on the first k coin tosses ($\mathcal{F}_k$), we should be able to tell if $\tau = k$ or not³. For this reason, stopping times have a very important property:

$$\text{if } \tau(\omega_1 \dots \omega_k \omega_{k+1} \dots \omega_N) = k, \text{ then } \tau(\omega_1 \dots \omega_k \omega'_{k+1} \dots \omega'_N) = k \text{ for all } \omega'_{k+1} \dots \omega'_N.$$

Example 2. The random variable τ defined below is a stopping time:

$$\tau(HH) = \infty, \quad \tau(HT) = 2, \quad \tau(TH) = \tau(TT) = 1. \quad (10)$$

On the other hand, the random variable ρ defined below is not a stopping time:

$$\rho(HH) = 0, \quad \rho(HT) = 0, \quad \rho(TH) = 1, \quad \rho(TT) = 2. \quad (11)$$

We mentioned that the value process of an American derivative is not a martingale, because the contract lose value after the optimal exercise time. But the discounted value process is a martingale before the optimal exercise time. To formally describe this, we need the notion of **stopped** process. Given a stopping time τ and a stochastic process $X = \{X_k\}$, we can define a new stochastic process $Y = \{Y_k\}$ that coincides with X completely at any time up to and including τ , and then just to be constant:

$$Y_k(\omega) = X_{k \wedge \tau(\omega)}(\omega), \text{ where } k \wedge \tau(\omega) = \min(k, \tau(\omega)). \quad (12)$$

Intuitively, for each $\omega \in \Omega$ (each N letter string of H 's and T 's), we have a "path" for values of X . (Imagine plot points $(k, X_k(\omega))$ on a XY -plane.) We freeze the path after time $\tau(\omega)$, so that Y is a constant afterward.

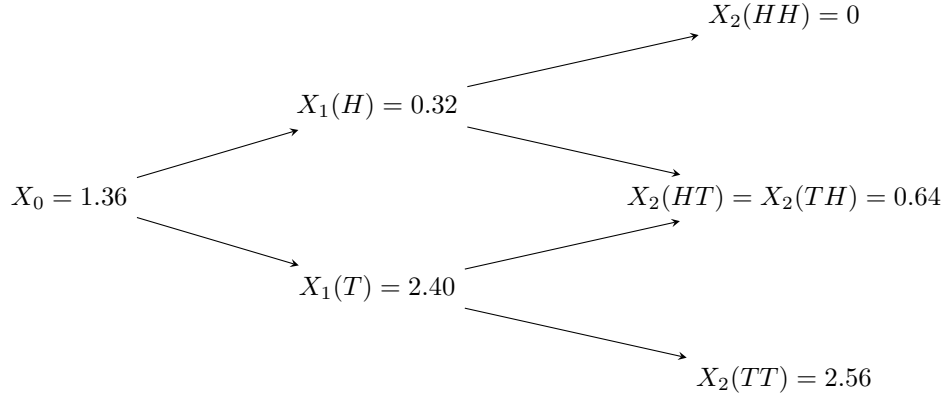


Figure 2: A two-period binomial model.

Example 3. In a two-period binomial model, let $X_k = \frac{1}{(1+r)^k} V_k$ be the discounted price of the American put we studied in *Example 1*. So that we obtain the following tree. Clearly X is not a martingale since

$$X_1(T) = 2.64 > \tilde{p}X_2(TH) + \tilde{q}X_2(TT) = \frac{1}{2} \cdot 0.64 + \frac{1}{2} \cdot 2.56.$$

However, using the stopping time introduced in *Example 2* to stop X , we have a new process, see Figure 3.

Stopped process persevere some properties of the original process. For example, if $X = \{X_k\}$ is a martingale (resp, supermartingale) for $k = 0, 1, \dots, N$, and τ is a stopping time, then the stopped process $\{X_{k \wedge \tau}\}$ is also a martingale (resp, supermartingale). This result is called the **optional sampling theorem**. In particular, the expectation of a martingale does not change if you stop it with a stopping time τ , since if X is a martingale, then ⁴

$$E(X_k) = X_0 = X_{0 \wedge \tau} = E(X_{0 \wedge \tau}) = E(X_{k \wedge \tau}). \quad (13)$$

²This is just a claim here. The complete proof will be given next time.

³A more rigorous way of stating it is: the event $\{\tau \leq k\} \in \mathcal{F}_k$

⁴We require X_0 to be $\mathcal{F}_0$ -measurable. Because $\mathcal{F}_0$ is just the trivial σ -algebra, X_0 must be a constant.

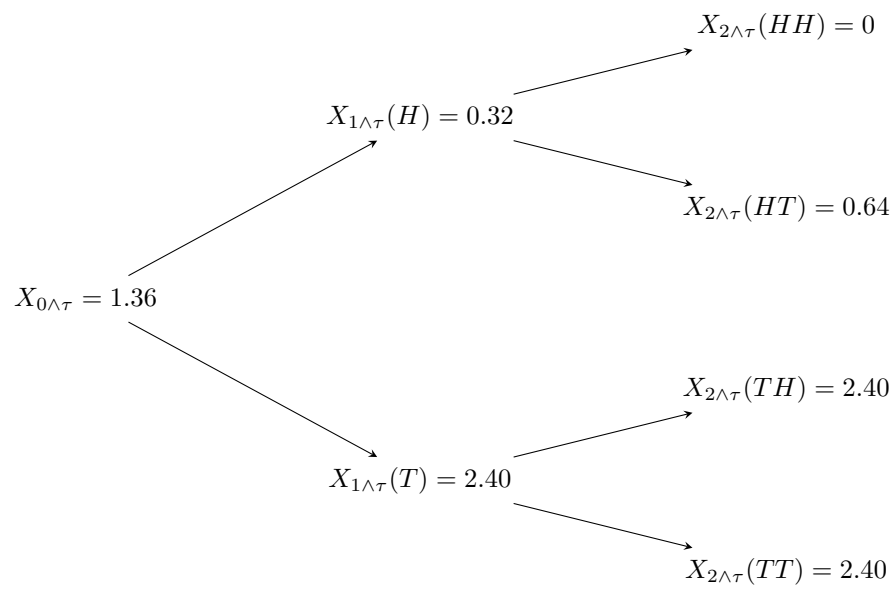


Figure 3: A stopped process.